

241901077

PONSURABHI V

Ex.No:13

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NETWORK FORENSICS USING PCAPXRAY TOOL

AIM:

To analyze a PCAP file using PcapXray and visualize network activity for basic network forensic investigation.

ALGORITHM:

1. Install PcapXray dependencies using: `sudo pip3 install -r requirements.txt`
2. Launch the tool with: `sudo python3 PcapXray.py`
3. Open the web interface and upload the given PCAP file.
4. Click Analyze to process the packet data.
5. View the generated network topology and communication graph.
6. Examine hosts, protocols, and suspicious connections.
7. Save the generated report and diagrams for documentation.

OUTPUT:

```
(JackSparrow@Captain)-[~/workspace]
$ sudo apt install python3 python3-pip python3-tk graphviz graphviz-dev tshark git -y
Note, selecting 'libgraphviz-dev' instead of 'graphviz-dev'
python3 is already the newest version (3.13.7-1).
python3 set to manually installed.
python3-pip is already the newest version (25.2+dfsg-1).
graphviz is already the newest version (2.42.4-3).
libgraphviz-dev is already the newest version (2.42.4-3).
git is already the newest version (1:2.51.0-1).
git set to manually installed.
The following packages were automatically installed and are no longer required:
  binutils-mingw-w64-i686      gcc-mingw-w64-x86-64-win32-runtime  mingw-w64-i686-dev
  binutils-mingw-w64-x86-64    libaio1t64                          mingw-w64-x86-64-dev
  gcc-mingw-w64-base           libwireshark18                      nasm
  gcc-mingw-w64-i686-win32     libwiretap15                        oracle-instantclient-basic
  gcc-mingw-w64-i686-win32-runtime  libwsutil16                        postgresql
  gcc-mingw-w64-x86-64-win32    mingw-w64-common                    ruby-json
Use 'sudo apt autoremove' to remove them.

(JackSparrow@Captain)-[~/workspace]
$ git clone https://github.com/Srinivas11789/PcapXray.git
Cloning into 'PcapXray'...
remote: Enumerating objects: 1704, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 1704 (delta 3), reused 9 (delta 2), pack-reused 1689 (from 1)
Receiving objects: 100% (1704/1704), 115.75 MiB | 5.48 MiB/s, done.
Resolving deltas: 100% (975/975), done.
```

```
(JackSparrow@Captain)-[~/workspace/PcapXray]
$ sudo apt install graphviz graphviz-dev python3-dev build-essential
Note, selecting 'libgraphviz-dev' instead of 'graphviz-dev'
graphviz is already the newest version (2.42.4-3).
libgraphviz-dev is already the newest version (2.42.4-3).
python3-dev is already the newest version (3.13.7-1).
python3-dev set to manually installed.
build-essential is already the newest version (12.12).
build-essential set to manually installed.
The following packages were automatically installed and are no longer required:
  binutils-mingw-w64-i686      libwiretap15
  binutils-mingw-w64-x86-64    libwsutil16
  gcc-mingw-w64-base           mingw-w64-common
  gcc-mingw-w64-i686-win32     mingw-w64-i686-dev
  gcc-mingw-w64-i686-win32-runtime mingw-w64-x86-64-dev
  gcc-mingw-w64-x86-64-win32   nasm
  gcc-mingw-w64-x86-64-win32-runtime oracle-instantclient-basic
  libaio1t64                  postgresql
  libwireshark18              ruby-json
Use 'sudo apt autoremove' to remove them.
```

Summary:

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 737

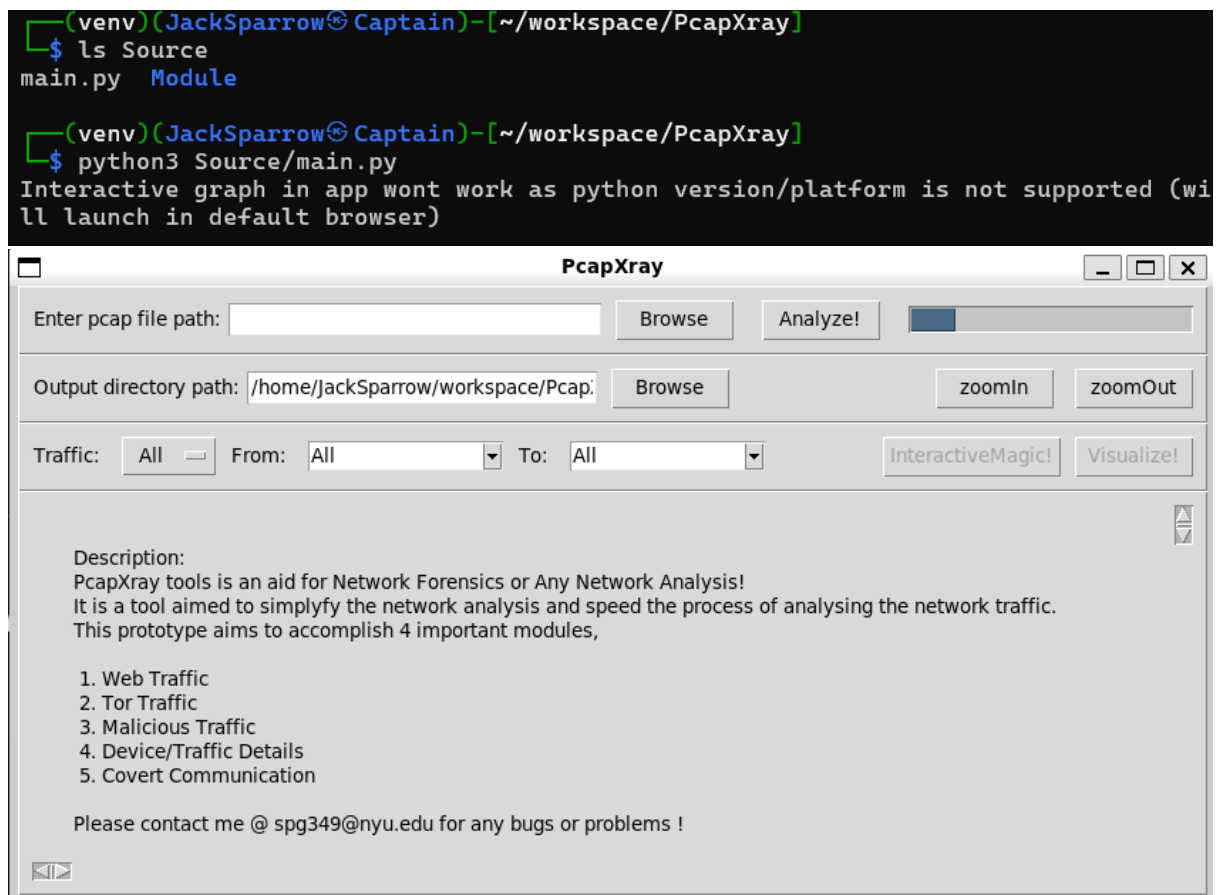
```
(JackSparrow@Captain)-[~/workspace/PcapXray]
$ python3 -m venv venv

(JackSparrow@Captain)-[~/workspace/PcapXray]
$ source venv/bin/activate

(venv)(JackSparrow@Captain)-[~/workspace/PcapXray]
$
```

```
(venv)(JackSparrow@Captain)-[~/workspace/PcapXray]
$ pip install pygraphviz
Collecting pygraphviz
  Downloading pygraphviz-1.14.tar.gz (106 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: pygraphviz
  Building wheel for pygraphviz (pyproject.toml) ... done
  Created wheel for pygraphviz: filename=pygraphviz-1.14-cp313-cp313-linux_x86_64.whl size=172936 sha256=316a0755305ced361018f49ebfb33737a9eb3e616b9ce25745a90947c16e2655
  Stored in directory: /home/JackSparrow/.cache/pip/wheels/20/62/e8/1c65bd9a08c9473ba98a15c4c7f62132ef090acac34b69a881
Successfully built pygraphviz
Installing collected packages: pygraphviz
Successfully installed pygraphviz-1.14
```

```
(venv)(JackSparrow@Captain)-[~/workspace/PcapXray]
$ pip install -r requirements.txt
Collecting scapy (from -r requirements.txt (line 4))
  Using cached scapy-2.6.1-py3-none-any.whl.metadata (5.6 kB)
Collecting pyshark (from -r requirements.txt (line 5))
  Using cached pyshark-0.6-py3-none-any.whl.metadata (806 bytes)
Collecting ipwhois (from -r requirements.txt (line 12))
  Using cached ipwhois-1.3.0-py2.py3-none-any.whl.metadata (21 kB)
Collecting netaddr (from -r requirements.txt (line 15))
  Using cached netaddr-1.3.0-py3-none-any.whl.metadata (5.0 kB)
Collecting pillow (from -r requirements.txt (line 18))
  Downloading pillow-12.0.0-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (8.8 kB)
```



RESULT:

Network forensics was successfully carried out using PcapXray, and the tool visualized hosts, protocols, and communication flows from the provided PCAP file.